

OLIVIA, MN, USA

WMO: 722168

Lat: 44.779N

Lon: 95.033W

Elev: 1076

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.0	-17.6	2.2	-10.1	-14.7	2.6	-7.6	32.2	11.3	28.5	14.1	7.4	320	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	89.9	73.5	85.8	70.7	81.8	68.4	75.8	84.7	73.4	81.8	71.6	79.0	12.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	125.5	81.8	71.5	121.1	80.1	69.6	113.1	77.8	39.9	85.3	37.7	81.6	36.1	79.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	23.7	20.3	DB	-19.4	94.6	4.5	2.8	-22.6	96.5	-25.3	98.1	-27.8	99.7	-31.1	101.7	(4)
(5)			WB	-19.8	80.1	4.5	2.2	-23.0	81.7	-25.6	83.0	-28.1	84.2	-31.3	85.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	15.5	17.0	31.7	44.7	57.1	67.2	71.1	67.5	61.6	47.4	34.3	19.6		
	DBStd	22.06	12.83	12.27	12.40	10.28	9.11	6.10	5.61	5.58	8.17	9.37	10.76	11.78		
	HDD50	4414	1068	923	580	214	35	0	0	0	8	163	480	943		
	HDD65	7924	1533	1343	1034	609	276	43	11	36	158	552	921	1408		
	CDD50	2479	0	0	11	56	255	516	653	542	356	82	8	0		
	CDD65	517	0	0	1	1	31	109	199	113	57	6	0	0		
	CDH74	4455	0	0	4	24	372	946	1707	838	495	68	1	0		
	CDH80	1176	0	0	0	4	115	262	482	178	122	13	0	0		
(14) Wind	WSAvg	9.2	10.3	10.1	10.1	10.9	10.3	8.5	6.7	6.2	8.0	9.4	9.9	9.8		
(15) Precipitation	PrecAvg	24.4	0.7	0.6	1.3	2.4	3.0	3.9	3.3	3.3	2.6	1.9	1.3	0.6		
	PrecMax	35.2	4.6	2.1	5.4	6.2	7.2	12.0	8.5	7.8	6.9	6.0	3.7	1.7		
	PrecMin	14.4	0.0	0.0	0.2	0.3	0.4	0.2	0.6	0.7	0.5	0.0	0.0	0.0		
	PrecStd	5.0	0.7	0.5	0.9	1.3	1.5	2.2	1.8	1.5	1.5	1.5	1.1	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	53.6	70.0	77.1	90.4	90.5	92.9	90.1	90.1	81.4	68.0	50.1		
		MCWB	37.0	49.1	58.6	59.0	67.3	71.9	79.0	76.3	71.5	62.3	55.4	46.0		
	2%	DB	36.7	42.6	60.9	72.1	81.8	87.8	89.8	84.0	82.2	73.5	59.4	41.1		
		MCWB	34.5	37.9	52.4	56.6	63.0	70.6	76.0	72.8	67.6	58.3	50.2	36.2		
	5%	DB	34.3	36.5	54.2	67.7	77.4	82.0	85.8	81.4	79.7	69.6	54.2	36.4		
		MCWB	33.1	34.1	49.3	53.7	60.9	67.0	72.8	69.7	65.7	57.4	46.5	34.3		
	10%	DB	32.3	34.0	49.6	61.2	72.9	79.3	81.7	78.9	76.6	62.6	50.0	34.2		
		MCWB	31.7	32.6	45.1	50.2	59.8	64.9	69.9	68.0	65.1	52.7	44.2	33.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	48.4	60.4	61.2	70.5	76.6	81.0	77.6	74.3	68.6	57.1	46.4		
		MCDB	42.3	53.1	68.1	74.6	83.4	87.7	89.4	88.0	84.4	74.9	65.0	48.6		
	2%	WB	34.7	38.2	53.4	58.3	67.0	72.6	77.2	74.4	71.6	62.3	50.9	37.3		
		MCDB	36.7	42.1	57.6	68.9	76.8	81.7	86.9	82.1	79.5	72.5	58.3	40.1		
	5%	WB	33.3	34.3	48.6	55.1	64.3	70.3	74.3	72.1	69.1	57.3	47.2	34.5		
		MCDB	34.2	36.1	54.5	64.5	74.5	79.4	82.6	78.6	75.5	65.7	52.9	36.1		
	10%	WB	31.6	32.3	43.9	51.9	60.8	67.7	72.2	70.4	66.5	54.0	43.8	32.9		
		MCDB	32.3	33.6	49.1	60.1	70.9	76.5	79.9	75.9	74.2	60.8	49.8	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	16.7	15.5	19.2	20.7	18.8	19.5	19.3	21.3	19.1	16.7	14.3		
		MCDBR	16.1	18.7	25.0	28.4	27.3	22.7	21.7	21.7	24.8	28.2	24.8	16.9		
		MCWBR	14.3	15.5	16.7	17.0	13.8	11.2	11.8	12.0	12.8	15.9	15.9	14.3		
	5% WB	MCDBR	15.4	17.4	23.8	24.7	22.4	19.3	19.4	18.5	20.5	22.9	23.0	15.1		
		MCWBR	13.9	14.8	16.5	15.9	13.1	11.6	11.6	12.0	12.8	14.4	15.6	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.277	0.306	0.363	0.384	0.395	0.401	0.408	0.366	0.318	0.280	0.259		
	taud		2.468	2.414	2.448	2.338	2.338	2.362	2.334	2.294	2.387	2.520	2.542	2.476		
	Ebn at Noon		279	292	296	285	281	277	274	267	271	273	269	267		
	Edh at Noon		22	28	31	37	39	38	39	39	33	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		574	906	1201	1487	1717	1912	2063	1705	1330	863	560	418		
	RadStd		65	105	118	144	131	139	71	136	98	116	51	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page